A Multimodal Deep Learning Framework for Non-Invasive Neonatal Jaundice (Bilirubin) Estimation Using Skin Images and Clinical Parameters

A

Project Report

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

by

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Jan – 2026 to May – 2026

CANDIDATE'S DECLARATION

We hereby certify that the project work entitled **"A Multimodal Deep Learning Framework for Non-Invasive Neonatal Jaundice (Bilirubin) Estimation Using Skin Images and Clinical Parameters"** submitted in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology** in **Computer Science and Engineering** to the AI Cluster, School of Computer Science, University of Petroleum & Energy Studies, Dehradun, is an authentic record of work carried out by us between **January 2026** and **May 2026** under the supervision of **Dr. Sanoj Kumar, Associate Professor, School of Computer Science, UPES, Dehradun.**

This project has not been submitted for the award of any other degree at this or any other university.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ABSTRACT

Approximately **60% of full-term** and **80% < full-term** babies have neonatal jaundice (hyperbilirubinemia). of low weight infants during the first week of life. The majority of the cases pass out of their own as the liver. grows up, though some of them may go on to develop kernicterus - brain damage caused by bilirubin, which occurs in some babies. is the cause of cerebral palsy, hearing loss and permanent impairment of the brain. The standard diagnostic test, Total Serum Bilirubin (TSB) test, involves a blood sample which is. emerges as a distressing factor to the neonate and creates laboratory turnaround delays. Transcutaneous bilirubinometers minimize patient pain, but are costly and lose their dependability at the high. bilirubin levels that are at the highest urgency level in making treatment decisions.

This project suggests and analyzes a **multimodal deep learning system that approximates**. non-invasively non-invasive neonatal bilirubin levels can be obtained in two sources: (i) calibrated high-resolution. images of the skin, colour features based on RGB and HSV colour spaces, and (ii) structured clinical parameters, including gestational age, postnatal age as well as birth weight, which are are routinely available in the clinical record, and are known to substantially affect bilirubin. dynamics. A personalized **2D Convolutional Neural Network (2D-CNN)** is used in the visual branch, which is operationalized. on denoised 567 x 567 Region of Interest (ROI) images. They are manufactured by means of a. pipeline of HSV-based colour card detection pipeline, LAB colour space white balancing pipeline, yellow- region detection through the use of the B channel, and morphological skin segmentation. The clinical branch uses a **Multilayer Perceptron (MLP)** to encode the three clinical covariates into a compact learned representation.

The two branches are trained together and their results combined via. concatenation at feature level followed by the ultimate bilirubin risk grouping. There were four model architectures that were assessed using the publicly available NeoJaundice dataset. There were (2,235 images, 745 neonates) and further cross-validation was performed on a personal clinical corpus. A 1D-CNN baseline was able to reach 76 percent accuracy; **2D-CNN that optimised preprocessing** reached **76 percent accuracy.** achieved 81.33% transfer learning; **ResNet-50 with transfer learning** achieved **72%** transfer learning; and ResNet-50 with transfer learning achieved 72% transfer learning. **with complete pipeline 78% was attained.** The main conclusion is that the quality of preprocessing - specifically colour calibration, skin ROI extraction - performance is more driven than otherwise. architecture choice does. Class imbalance 81% majority class was solved by use of. stratified splitting and inverse-frequency weighting of classes.

The completed multimodal system has potential as a practical, low-cost bilirubin screening

tool for resource-limited clinical settings where laboratory infrastructure is unavailable and every blood draw has real cost to the neonate.

Keywords: Neonatal Jaundice, Bilirubin Estimation, 2D-CNN, Multimodal Deep Learning, Skin Colour Analysis, Non-Invasive Screening, Clinical Parameter Fusion, Total Serum Bilirubin, ROI Extraction, Transfer Learning, Class Imbalance.

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CHAPTER 1

INTRODUCTION

1.1 Background

Walk into almost any neonatal ward and you will find nurses holding newborns up to the window light, squinting at the shade of yellow spreading across tiny foreheads and chests. Jaundice is that common. Neonatal hyperbilirubinemia — the technical name for abnormally elevated bilirubin in a newborn's blood — affects roughly **60% of full-term** and up to **80% of preterm**

In the case of the majority it is self-limiting. when the liver develops and the bilirubin clearance is increased. But in a worrying minority, in case of the condition is overlooked or underestimated, the aftermath is disastrous and irreversible. Bilirubin, in excess of a certain level, diffuses through the blood-brain barrier. and foci in the basal ganglia and brainstem, which result in kernicterus - a form of bilirubin- caused brain injury which is associated with cerebral palsy, hearing loss, cognitive impairment, and developmental delay.

In low income nations where laboratory facilities are few in between. and clinical follow-up is difficult to maintain, **kernicterus** still is one of the most widespread etiologies. of avoidable inborn illness in babies. It is not the life of the condition that is tragic but the. reason that the timely diagnosis and treatment of diseases are virtually always possible, provided that these diseases are diagnosed early and treated in time. transfusion which averts the worst results altogether.

1.2 Motivation

The existing gold standard of diagnosing neonatal jaundice is the Total Serum Bilirubin. A heel-prick or a venipuncture blood taking is required as part of (TSB) test. While technically this is a painful process that is distressing to the neonate, and consumes nursing time, producing. laboratory costs, and brings about turnaround delays within which bilirubin may persist. to rise. Repeat blood sampling of due to repeated blood sampling: In a busy district hospital where fifty neonates were being seen each day, the repeated blood sampling of a due was undergoing repeated blood sampling. The jaundice monitoring are actually cumbersome.

To allow this burden to be reduced, transcutaneous bilirubinometers (TCB devices) were developed. They are quick, non-invasive and fairly accurate even in mild cases. They have

a per-unit, however. cost such that they are not affordable to most district-level and rural hospitals. More critically, TCB values become inaccurate at bilirubin levels over 15mg/dL - exactly the level of bilirubin. scope in which treatment decisions are most in need.

Those restrictions have spurred an increasing amount of literature in camera-based, machine learning- powered bilirubin estimation. The argument is clear: bilirubin is a yellow stain on the skin that is caused by a. Progression in cephalocaudal which is associated with serum concentration within a clinically relevant. range. This yellowing can be seen in a photograph, and can move in colour channels in ways. that have quantitative information. Then arises the question: how do we, so that we are able to extract that. information in a reliable manner and what do we couple it with the clinical context which identifies what. a specific bilirubin level really implies to the particular baby?

1.3 Problem Statement

No existing non-invasive bilirubin estimation method satisfies all practical requirements at once. Blood tests are accurate but painful and slow. TCB devices are gentle but costly and unreliable at dangerous bilirubin levels. Existing image-based deep learning systems have demonstrated that skin yellowing carries learnable bilirubin information, but virtually all of them treat the problem as purely visual, discarding the clinical metadata — gestational age, postnatal age, birth weight — that every neonatologist considers when assessing the same infant. This is a fundamental mismatch between what the model knows and what a clinician needs. This project is designed to close that gap.

1.4 Original Objectives (Initial 1D-CNN Approach)

The project was initially conceived around a 1D-CNN architecture that treated skin colour values as sequential signals. The objectives at that stage were as follows.

Main Objective: To develop and validate a non-invasive multimodal deep learning framework that integrates neonatal skin image features and clinical parameters for accurate bilirubin level estimation and early jaundice screening.

- **Extract and analyse skin colour features** from infant images across multiple colour spaces and model these features using a **1D Convolutional Neural Network (1D-CNN)**.
- **Process relevant clinical parameters** through a neural network–based approach and fuse them with visual features for improved bilirubin prediction accuracy.
- **Evaluate the proposed framework** using standard regression metrics and assess its effectiveness as a screening tool for early neonatal jaundice detection.

1.5 Revised Objectives (Final 2D-CNN Multimodal Approach)

After the 1D-CNN baseline experiments, the architectural limitations of treating spatial skin images as 1D sequences became clear. Variable sequence lengths caused inconsistent gradient flow, and the 1D representation discarded spatially structured colour information that turned out to matter. The objectives were therefore revised to reflect a 2D-CNN multimodal framework with an explicit preprocessing pipeline as a first-class component.

Main Objective: To develop and validate a non-invasive multimodal deep learning framework that integrates neonatal skin image features and structured clinical parameters — gestational age, postnatal age, and birth weight — for accurate bilirubin level estimation and early jaundice risk classification.

- **Develop a robust image preprocessing pipeline** encompassing colour calibration (HSV-based colour card detection, LAB white balancing), yellow-region detection via the B channel, skin segmentation, and ROI extraction that produces consistent, device-independent skin colour representations from neonatal photographs.

- **Design and implement a custom 2D-CNN visual branch** that learns spatial and colour-based features associated with bilirubin concentration from high-resolution preprocessed ROI images, without relying on hand-crafted feature engineering.

- **Design and implement a clinical MLP branch** which encodes gestational age, postnatal age, and encode the inputs of birth weight and weight into a small learned network, which represents the physiological risk. modulating factors that influence the bilirubin dynamics and treatment levels.

- **Combine the visual and clinical branch at the feature level** and train the entire. end-to-end multimodal system, which allows common reasoning on visual or clinical data.

- **Systematically compare four model architecture** (1D-CNN, 2D-CNN, EfficientNet-B0, To gain insight into ResNet-50) on the public NeoJaundice dataset and a private clinical corpus to understand. how the performance varies with the improvement of architecture and preprocessing quality.

- **Assess the final structure** based on classification measures (accuracy, precision, recall, F1 score) Evaluate the final structure based on classification measures (accuracy, precision, recall, F1 score, ROC-AUC) and regression statistics (RMSE, R2), also taking into account the imbalance in classes. manipulation by stratified splitting and inverse-frequency weighting of the class.

1.6 Project Timeline (PERT Chart)

Figure 1.1 shows the project schedule as a PERT chart. The project ran for approximately 18 weeks from January to May 2026 and was divided into five phases: planning and literature review, dataset acquisition and exploratory analysis, preprocessing and colour calibration, model design and training, and results analysis and report writing.

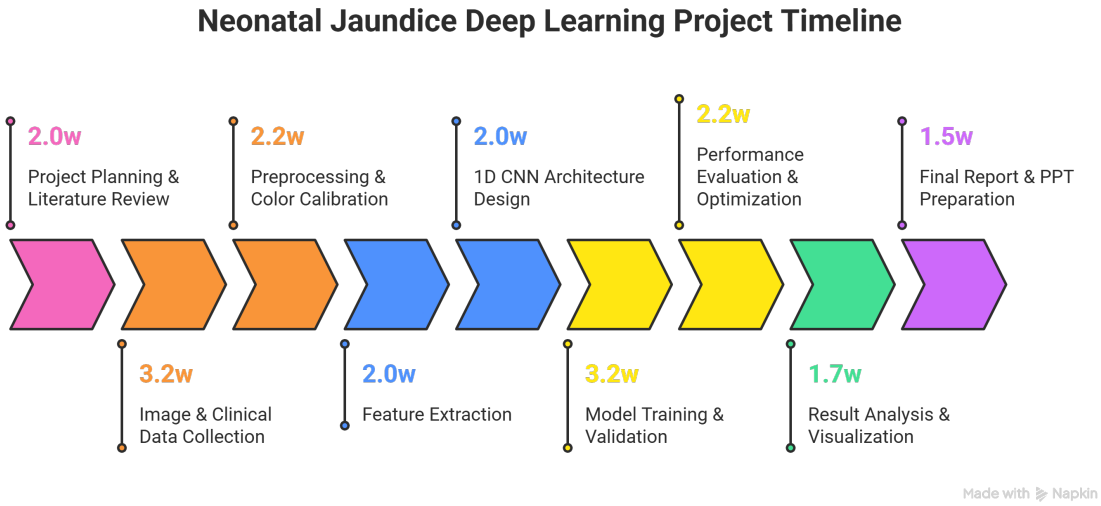


Figure 1.1: PERT Chart for the Neonatal Jaundice Deep Learning Project (January–May 2026)

1.7 Report Organisation

Chapter 2 involves the system analysis that involves existing approaches, limitations of the approaches, and so on. the proposed architecture. Chapter 3 contains the description of system design: the preprocessing pipeline, data flow, model architectures, and data flow. The datasets are depicted in Chapter 4. Chapter 5 covers implementation. The 6th chapter provides and explains the results of experiments. Chapter 7 addresses constrained work and work in future. Chapter 8 concludes.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Existing Systems and Literature Review

Studies regarding non-invasive methods of detection of jaundice have come a long way in the last. decade, shifting away simple colour-index formulas used upon digital photographs to complete. deep learning pipelines that use high resolution smartphone images. Understanding what the preceding methods get right and where they go wrong it was the point of departure of our. design

2.1.1 Image-Based and Machine Learning Approaches

The common principle underlying most of the current systems is the fact that bilirubin stains the. skin yellow and that yellowing can be determined by a photograph. Hardalac et al. used support vector regression on colour values sampled on several are used as k-nearest neighbours. body parts with an accuracy of 85-92 percent with a smartphone camera. BiliCam combined an ensemble of five regression algorithms with a physical colour calibration card for home monitoring. Mansor *et al.* used skin hue analysis across facial and body regions, exceeding 90% accuracy under controlled laboratory conditions.

These classical approaches showed the concept was viable but required manual feature engineering and calibration hardware. They were also evaluated on small, demographically narrow datasets, which left the generalisation question open.

2.1.2 Deep Learning Approaches

The more recent generation of approaches feeds skin images directly into convolutional networks, bypassing hand-crafted features. Makhloughi converted skin ROI pixel values into 1D sequential representations and applied a 1D-CNN, achieving bilirubin regression without any manual features. Chakraborty *et al.* combined a 2D-CNN with the Segment Anything Model and CLAHE preprocessing and added XAI explanations — at the cost of accuracy (0.78) and with a requirement for uncalibrated images. Salami *et al.* surveyed 51 studies and found consistent accuracy above 90% in controlled settings but persistent problems with lighting sensitivity and skin-tone bias across the literature.

Table 2.1: Key Limitations of Existing Non-Invasive Jaundice Detection Systems

Limitation	Description
Clinical data ignored	Almost all image-based models ignore gestational age, postnatal age, and birth weight — variables that directly modulate bilirubin dynamics and treatment thresholds.
Lighting/device sensitivity	Without colour calibration, the same baby can appear very differently across devices and lighting environments, causing model failure on out-of-distribution images.
Class imbalance ignored	Datasets with 70–80% jaundice-positive samples allow naive models to achieve high accuracy by predicting the majority class for every input.
Small or synthetic datasets	Many studies use small datasets ($<$100 samples) or internet/3D-rendered images, limiting real-world applicability.
Single-modality only	Pure vision systems cannot recover clinically important context from pixel values alone; a premature baby and a term baby with identical skin colour require very different management.
Interpretability gaps	Most high-performing DL models provide no explanation of their predictions, which is a barrier to clinical adoption.

2.1.3 Limitations of Existing Systems

2.2 Motivations for the Proposed System

- **Clinical data fusion:** A neonatologist assessing a jaundiced baby never looks at the skin alone. Gestational age, postnatal age, and birth weight are the primary axes on which every jaundice management protocol is built. Any model that ignores them is solving an incomplete problem.
- **Colour calibration:** The NeoJaundice dataset was collected with a reference colour card precisely to enable post-hoc calibration. Using this information removes the device-dependent and illumination-dependent colour shifts that have undermined many earlier approaches.
- **Class imbalance handling:** The 81% majority-class prevalence in the primary dataset is a real threat to the validity of raw accuracy as a metric. Explicit class weighting is not optional — it is a prerequisite for meaningful evaluation.
- **Affordable deployment target:** The system is designed around images that could be captured by a standard smartphone with a physical reference card, requiring no specialist hardware.

2.3 Proposed System Overview

The proposed system is a multimodal deep learning framework with two parallel branches whose learned representations are merged at the feature level before the final output. Figure 2.1 shows the high-level architecture.

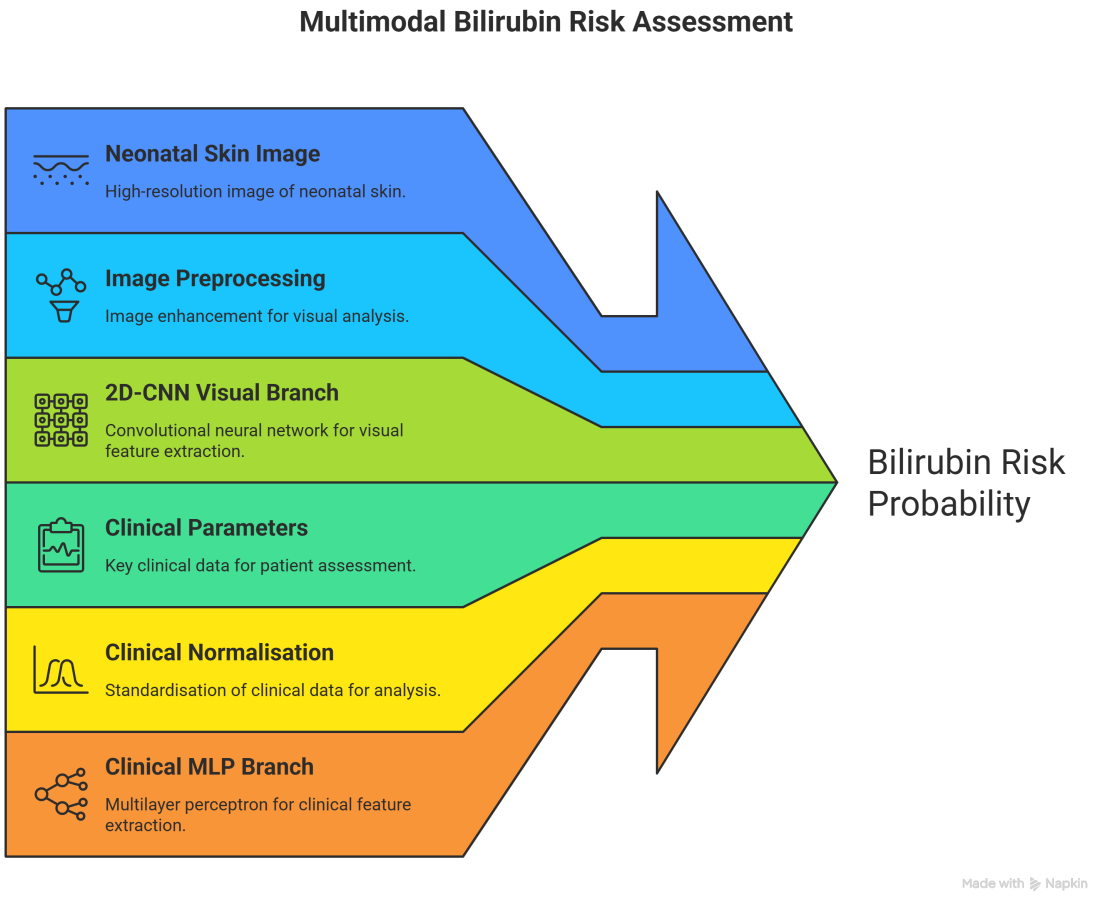


Figure 2.1: High-level architecture of the proposed multimodal neonatal bilirubin estimation system

2.4 System Modules

2.4.1 Image Preprocessing Module

Handles colour calibration, spatial normalisation, yellow-region detection, and skin segmentation. This module transforms raw neonatal photographs into calibrated, masked skin ROI patches suitable for convolutional feature learning.

2.4.2 Visual Feature Learning Module (2D-CNN)

A three-block 2D convolutional network with increasing filter depth ($16 \rightarrow 32 \rightarrow 64$) that learns spatial and colour-based bilirubin-related features from high-resolution ROI images without manual feature engineering.

2.4.3 Clinical Feature Encoding Module (MLP)

A compact two-hidden-layer MLP that encodes the three clinical covariates (gestational age, postnatal age, birth weight) after z-score normalisation into a 16-dimensional embedding.

2.4.4 Multimodal Fusion and Classification Module

Concatenates the 64-dimensional visual embedding from the 2D-CNN FC head with the 16-dimensional clinical embedding and passes the combined 80-dimensional vector through a sigmoid output unit to produce the bilirubin risk probability.

CHAPTER 3

SYSTEM DESIGN

3.1 Design Philosophy

The design of this system rests on one principle: the preprocessing pipeline is not preliminary to the model — it is part of the model. Our experiments showed repeatedly that input representation quality has a larger effect on downstream performance than architecture choice. Design decisions were made preprocessing-first and architecture-second.

3.2 Preprocessing Pipeline Design

Figure 3.1 shows the detailed preprocessing pipeline for a sample NeoJaundice image (`0001-1.jpg`). Starting from the original 567×567 native image (panel 1), the pipeline detects the yellow reference patch from the colour card for white balance reference (panel 2), generates a skin mask using HSV thresholding (panel 3), applies LAB white balancing to produce a colour-normalised image (panel 4), extracts the masked skin pixels (panel 5), and finally serialises the 72,676 non-zero skin pixels into a 4096×6 RGB+HSV feature sequence for the 1D-CNN baseline or a resized 2D patch for the CNN branches (panel 6).

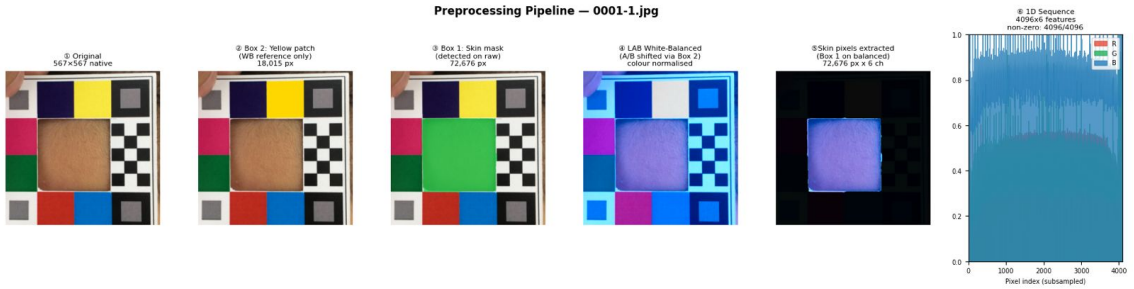


Figure 3.1: Image preprocessing pipeline (step-by-step) on `0001-1.jpg`: (1) original. 567×567 image, (2) yellow patch detection for WB reference, (3) skin mask (72,676 px), (4) LAB white-balanced output, (5) masked skin pixels extracted, (6) 1D RGB+HSV feature. sequence.

3.3 2D-CNN Architecture Design

The 2D-CNN visual branch was developed based on four principles: Progressive depth:

- **Number of filters** doubles across blocks (16 32 64) and the filter counts increase accordingly (e.g. 4000). network to learn more abstract colour and texture representations.

- **Spatial compression:** 2×2 max-pooling at the end of each convolutional block has the effect of reducing spatial. one-half, and so on, moving the effective receptive field along.
- **Certain parameters are regularised:** Dropout (0.1 per block) will not allow feature detectors to co-adapt.
- **Compact FC head:** The flattened feature volume is mapped by two dense layers (128 and 64 units). into a small bilirubin-responsible form.

3.4 MLP Clinical Branch Design

The clinical MLP is purposely small, as there are not many input features. The network takes as input a 3-dimensional z-score-normalised input vector and processes it with. two hidden layers (32 units, followed by 16 units, both ReLU with Dropout 0.2) that generate a. 16-dimensional clinical embedding which is concatenated with the CNN visual features at the fusion stage.

3.5 Data Flow Diagram

Figure 3.2 shows the conceptual preprocessing flow, and Figure 3.3 represents the entire end-to-end flow of data of raw inputs. via both branches to the end sigmoid output.

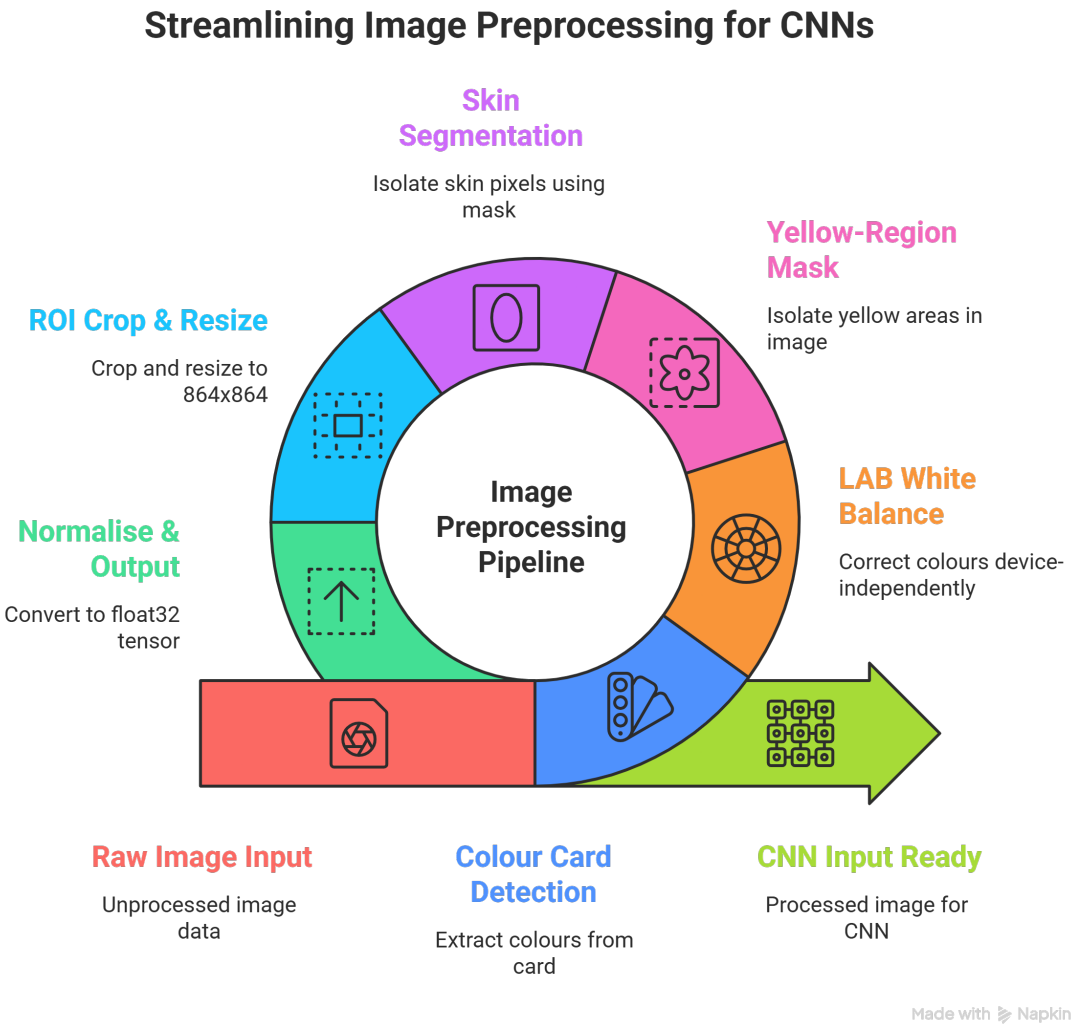


Figure 3.2: CNN input conceptual image preprocessing pipeline.

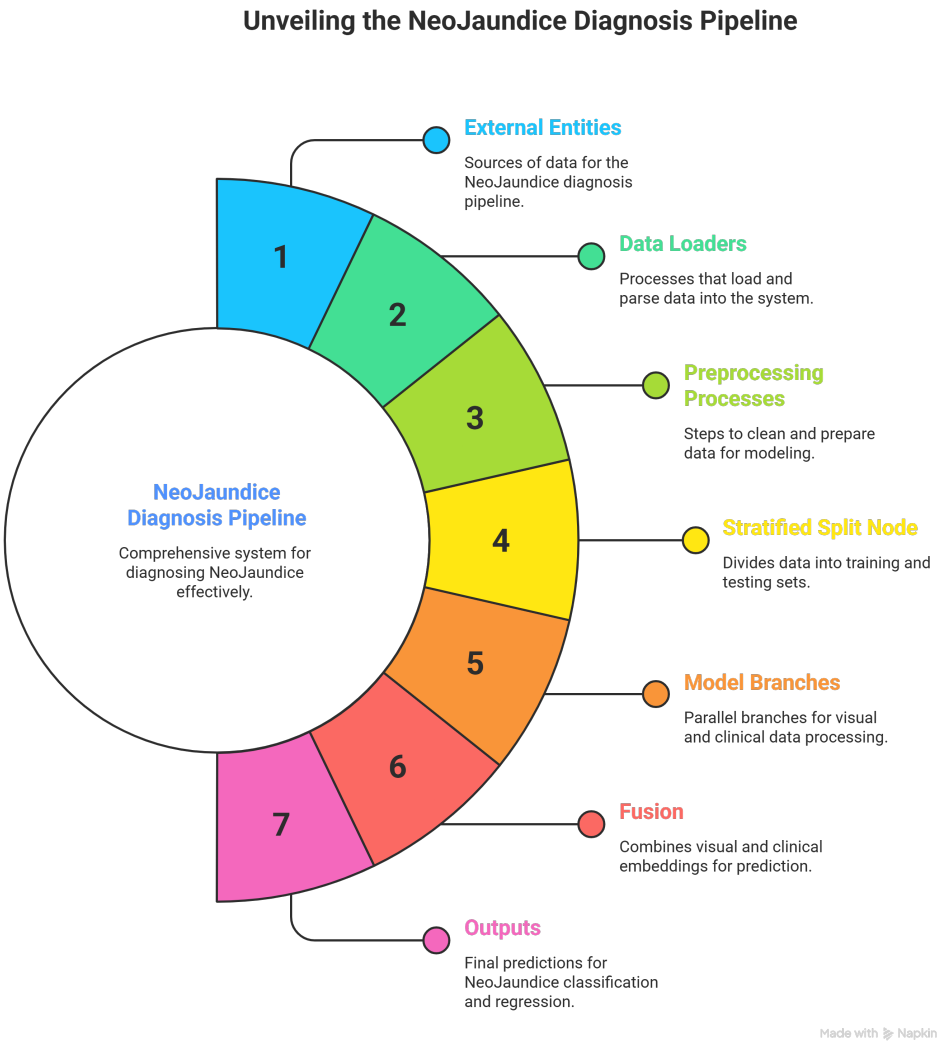


Figure 3.3: Multimodal framework End-to-end data flow diagram of the multimodal framework.

CHAPTER 4

DATASET DESCRIPTION

4.1 Primary Dataset: NeoJaundice

The main data set of all the training and evaluation experiments is the publicly. accessible NeoJaundice dataset, available on the SpringerNature Figshare. repository. It is among the most well documented series of neonatal. images of skins that can be researched.

4.1.1 Dataset Composition

- **2,235 high-resolution skin images** of 745 infants at birth, three images per. neonate that has been captured on the head (forehead) and chest areas.
- Each photograph was photographed along with a standardised reference colour card, allowing. post-hoc colour calibration.
- Images are jpeg format with their native size being 567 x 567 pixels as it is verified by the preprocessing pipeline.

4.1.2 Associated Clinical and Label Data

For all entries found in the provided CSV file, we have:

- **image_idx:** Name of the image that corresponds to each data entry.
- **gestational_age:** gestational age in weeks (range: 29–42).
- **age_day:** Age of the infant in days after birth (0-35).
- **weight:** Weight at birth in grams (1,800-5,000).
- **TSB:** Total Serum Bilirubin in mg/dL (range: 0.1–25.5).
- **Treatment:** binary label (0 = no phototherapy required; 1 = phototherapy required).

4.1.3 Data Distribution Analysis

The most clinically important characteristic of the dataset is its severe class imbalance: approximately 81% of samples carry the phototherapy-required label. This is not a data collection artefact — it reflects the actual prevalence of significant jaundice among infants

Table 4.1: Statistical Summary of NeoJaundice Dataset Variables

Variable	**Min**	**Mean**	**Max**	**Std Dev**
Gestational Age (weeks)	29	38.6	42	2.1
Postnatal Age (days)	0	5.3	35	4.8
Birth Weight (g)	1800	3140	5000	460
TSB (mg/dL)	0.1	11.2	25.5	4.8
Treatment label = 1 (%)	$\approx$ **81%**			

presenting to the neonatal unit where the data were collected. A naive model that predicts the majority class for every input achieves 81% accuracy while being clinically useless. Any valid evaluation must account for this; any training procedure that ignores it will produce a mode-collapsed model.

4.2 Secondary Dataset: Private Clinical Corpus

The next stage involves performing a threshold on the B channel of LAB to create a binary mask that can detect areas that contain jaundice skin. Opening and closing operations are done to eliminate any falsely detected areas from the mask. The non-skin pixels are masked out with zero values.

CHAPTER 5

IMPLEMENTATION

5.1 Technology Stack

Table 5.1: Technology Stack Used in This Project

Component	**Technology / Version**
Programming Language	Python 3.10
Deep Learning Framework	TensorFlow 2.x / Keras
Computer Vision	OpenCV 4.x
Data Manipulation	Pandas 2.x, NumPy 1.26
Machine Learning Utilities	scikit-learn 1.4
Visualisation	Matplotlib 3.8, TensorBoard
Development Environment	Jupyter Notebook, VS Code
Hardware	Google Colab GPU
Dataset Storage	Local SSD + Google Drive

5.2 Data Loading and Splitting

The CSV file is loaded with Pandas and filtered to rows where the corresponding image exists on disk. The data is split 80/20 train-test with stratification on the `Treatment` column to preserve the natural class ratio in both splits.

Class weights are computed from the training split using the inverse-frequency formula and passed to the training procedure. This assigns a weight of approximately 3.5 to the minority (no treatment) class and 0.62 to the majority (treatment required) class.

5.3 Image Preprocessing Implementation

The preprocessing pipeline runs as a `tf.data` mapping function applied in parallel across all images during training.

5.3.1 Colour Calibration

Colour calibration is performed using OpenCV offline before model training. The reference colour card is detected using HSV thresholding on the known hue ranges of the card's colour

patches. A correction matrix is computed in the CIE LAB colour space by mapping the detected card colours to their reference values, then applied to the full image via an affine transform in LAB space.

5.3.2 Yellow-Region Detection

After white balancing, the LAB B channel (the blue-yellow opponent axis) is thresholded to produce an initial binary mask of candidate jaundice-affected skin regions. Morphological opening removes spurious small detections while closing gaps in true skin regions. The refined mask zeroes out all non-skin pixels in the calibrated image.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Overview of Experiments

Four different models have been built with the help of the NeoJaundice dataset. They are all different model designs, one being an iteration of the design process. It is inappropriate to showcase only one of the highest performing models as this will give us a false idea of not having iterations at all.

6.2 Model 1: 1D-CNN Baseline

In model 1, it is assumed that the pixels in the skin form time series signal and thus the combination of the RGB channel and HSV channel of the skin ROI are combined into one-dimensional signals, which will then be inputted into one-dimensional CNN with three convolutional layers (32, 64, 128 filters).

6.2.1 1D-CNN Regression Results

depicts the plots for the training phase of the regression model and its predictive capability. The plots in the figures include the following: (upper left) Training and Validation Losses (MSE) with a fast decrease then stabilization; (upper right) Mean Absolute Error per epoch between the two data sets; (lower left) Prediction vs. Actuals with $R^2 = 0.546$ showing some correlation with the actual TSB; and (lower right) Heteroskedastic residuals with high bilirubin levels.

6.2.2 1D-CNN Classification Results

Figure 6.2 displays the plot of classification loss, accuracy, and confusion matrix for three classes. The final validation accuracy is estimated at around 74.4%. From the confusion matrix, it is clear that the classifier does very well for the high bilirubin class (>10 mg/dL) concerning recall (241 out of 242), however, poorly in the medium class range between 5-10 mg/dL where misclassification is more significant, with only 41 out of 105 samples classified.

Conclusion: 76% can be considered deceptive because the classifier result depended on the dataset imbalance and not feature learning. Uneven gradient magnitude across batches was

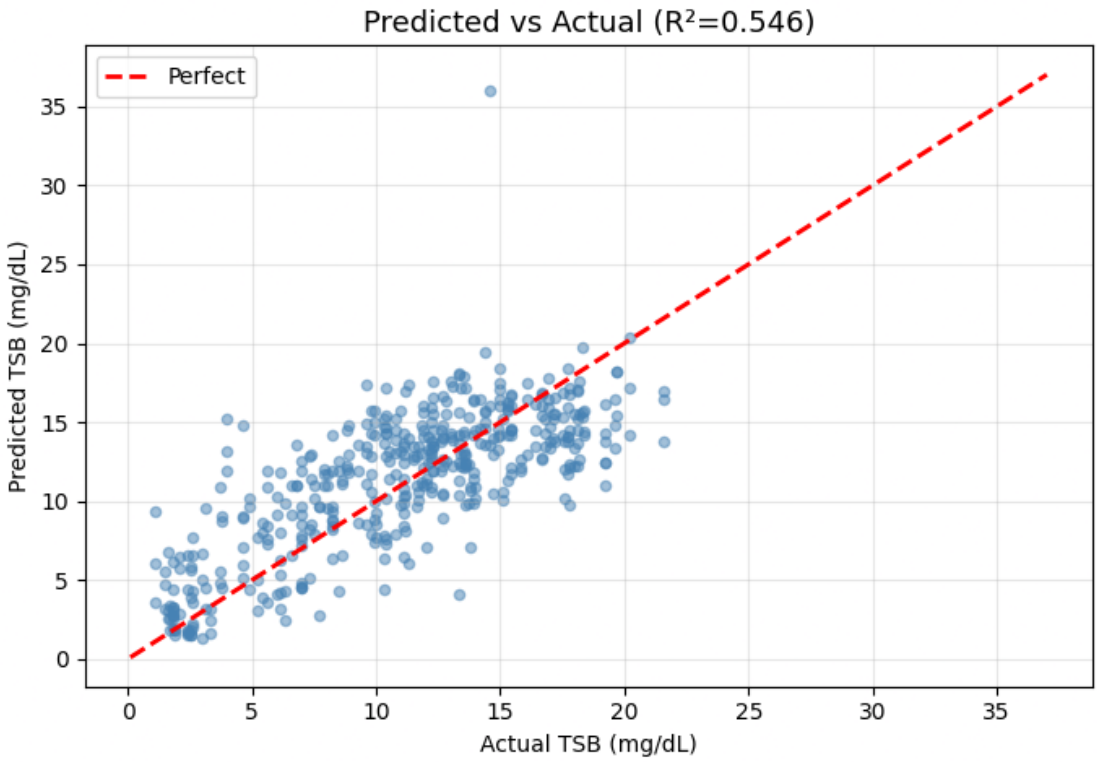


Figure 6.1: 1D-CNN Regression results: predicted vs. actual TSB ($R^2 = 0.546$).

noticed during the TensorBoard visualization, which was due to uneven effective sequence lengths owing to skin masks having unequal densities.

6.3 Model 1 (Fused): 1D-CNN + MLP Clinical Branch

To test the effect of fusing clinical parameters on the performance of 1D-CNN, an MLP component of the proposed architecture was included using the gestational age, age at birth, and birth weight.

6.3.1 Training Convergence

Figure 6.3 The following figure shows the graphs of loss function and mean absolute error of the hybrid model during the training process for 100 epochs. As seen, the similar trends in training and validation processes prove that overfitting did not occur in this case; otherwise, if the clinical parameters were useless, then overfitting was likely.

6.3.2 Multimodal Fusion Full Results

Figure 6.4shows the scatter plot of TSB prediction using the 1D-CNN multimodal fusion algorithm. The R-square value is $R^2 = 0.633$ and the mean absolute error is MAE = 2.58 mg/dL. The R^2 value has been improved from 0.546 (considering image features only) to

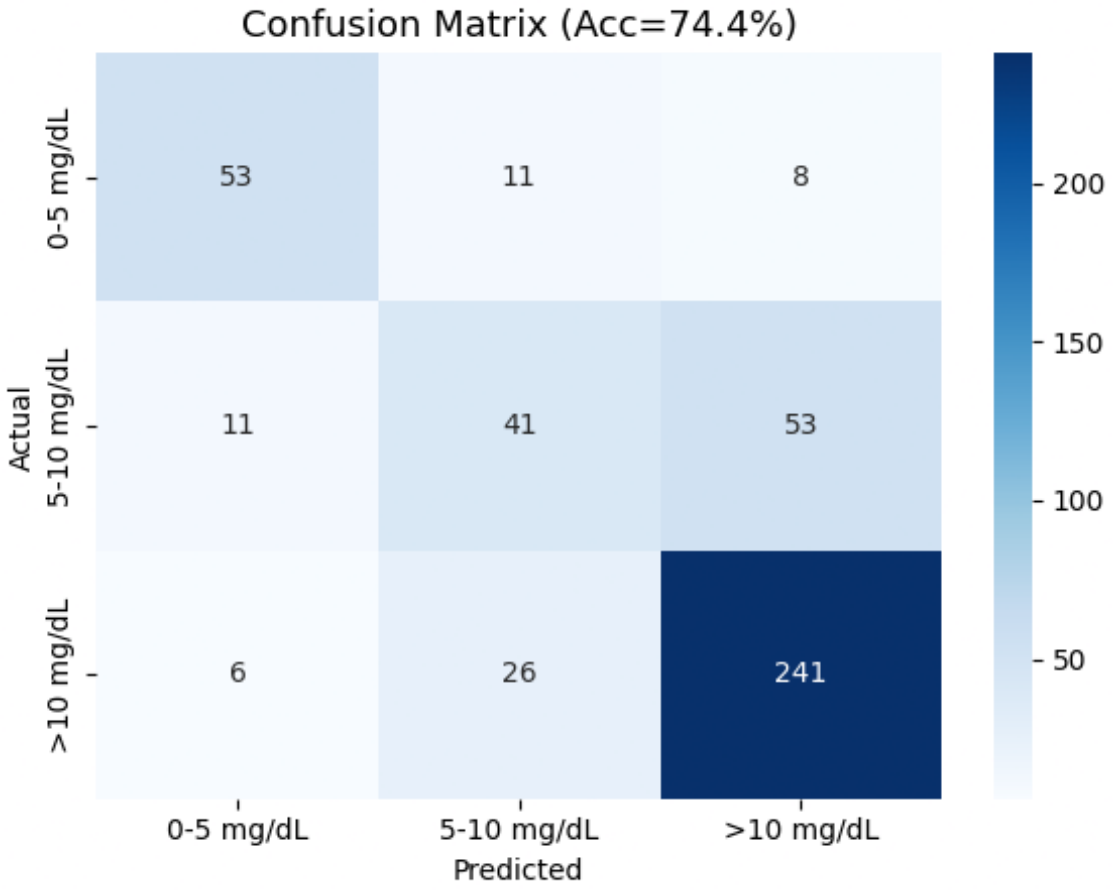


Figure 6.2: Results for Classification using 1D-CNN: (left) classification loss per epoch, (middle) classification accuracy per epoch, (right) confusion matrix (Accuracy = 74.4%) of three classes of bilirubin risk levels (0-5, 5-10, >10 mg/dL).

0.633. This demonstrates that gestational age, postnatal age, and birth weight offer some

6.3.3 Risk Classification Confusion Matrix

Figure 6.5 provides the risk level classification confusion matrix for the 1D-CNN fusion model that is being proposed. Recall of the perfect value obtained for the Low (226/228) and High (120/120) classes demonstrates a considerable impact of the clinical fusion on classification for minorities. Errors in classification that are associated with the Medium class (21 High) are the most difficult ones to explain because of uncertainties in clinical judgment in the 5-10 mg/dL range.

6.3.4 Best Fusion Run ($R^2 = 0.677$)

Figure 6.6 indicates the scatter plot and confusion matrix of the best performing 2D-CNN fusion model, where $R^2 = 0.677$ and RMSE = 2.970. Compared to Figure 6.4, the improvement in scatter tightness around the perfect-prediction line is clearly visible.

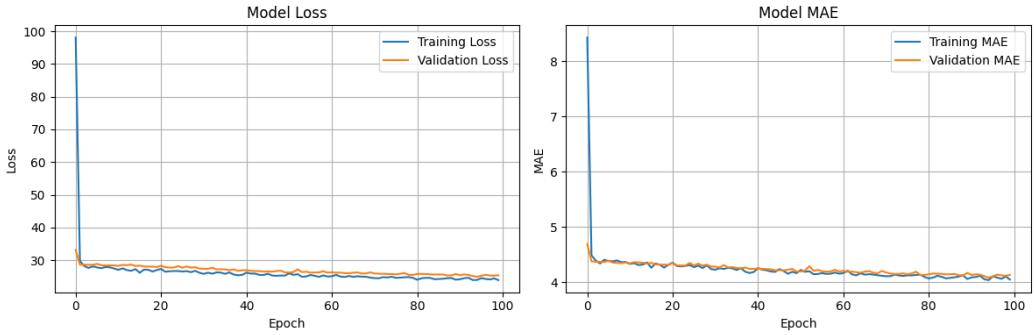


Figure 6.3: 1D-CNN + MLP fusion model: (left) training and validation loss (MSE) over 100 epochs, (right) training and validation MAE over 100 epochs.

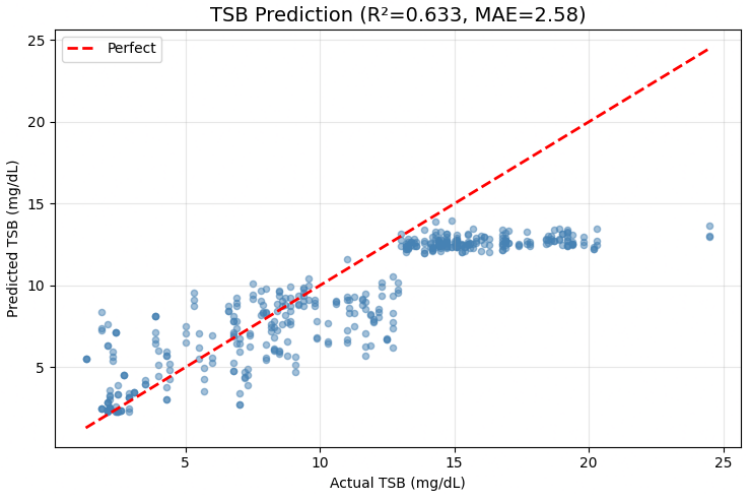


Figure 6.4: 1D-CNN + MLP Multimodal Fusion: TSB prediction scatter ($R^2 = 0.633$, MAE = 2.58 mg/dL).

6.4 Model 2: 2D-CNN with Optimised Preprocessing

The 2D-CNN architecture operated on consistently preprocessed 567 × 567 ROI images with colour calibration and explicit class weight balancing. **Result: 81.33% accuracy.**

Two failure modes were identified and resolved during development:

- **Mode collapse at low learning rate:** With an initial learning rate of 10^{-5}, the model rapidly converged to predicting 'treatment required' for all inputs. Raising the learning rate to 10^{-4} and applying class weights caused the model to genuinely discriminate between classes.

- **Imbalance-inflated accuracy:** Even after correction, raw accuracy is misleading at 81% majority-class prevalence. F1-score and AUC are the appropriate primary metrics.

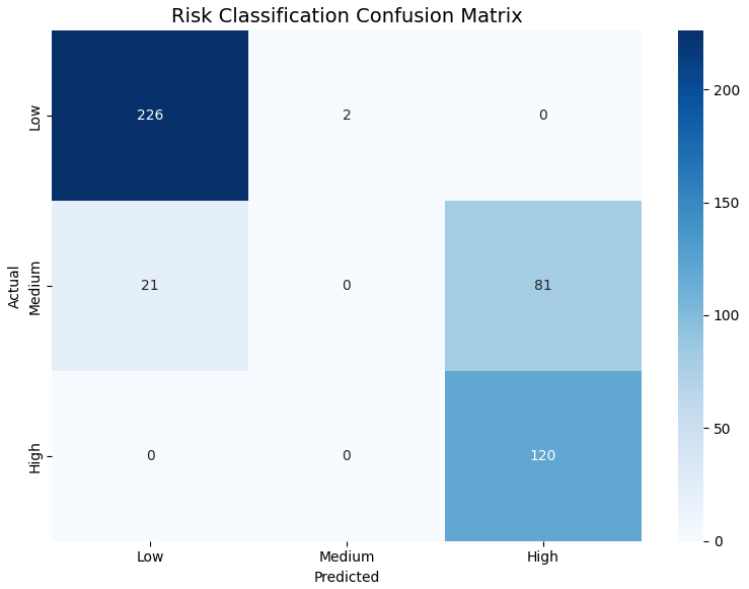


Figure 6.5: Risk classification confusion matrix for the 1D-CNN + MLP fusion model. Low-risk (226/228) and High-risk (120/120) classes achieve near-perfect recall; Medium-risk is the most difficult category.

6.4.1 Residual Analysis

Figure 6.7 shows the residuals distribution and residuals-vs-true-TSB scatter. The residual histogram is centred near zero, suggesting no systematic bias. The residuals-vs-TSB scatter reveals heteroscedasticity at higher TSB values — predictions become increasingly compressed below the true value beyond 15 mg/dL — a known challenge for regression models in this domain.

6.5 Model 2 (Fused): 2D-CNN + MLP 5-Fold Cross-Validation

To rigorously assess generalisation, the 2D-CNN multimodal fusion model was evaluated with 5-fold cross-validation. Figure 6.8 shows the training vs. validation loss curves for all five folds.

The consistent convergence across all five folds confirms that the model's performance generalises reliably across different data partitions rather than being specific to one split. This model also showed meaningful cross-dataset transfer to the private clinical corpus — an outcome attributed directly to the LAB white balancing step.

6.6 Model 3: EfficientNet-B0 with Transfer Learning

EfficientNet-B0 pretrained on ImageNet was fine-tuned with a custom classification head (Global Average Pooling, Dense 128 ReLU, Dense 1 Sigmoid).

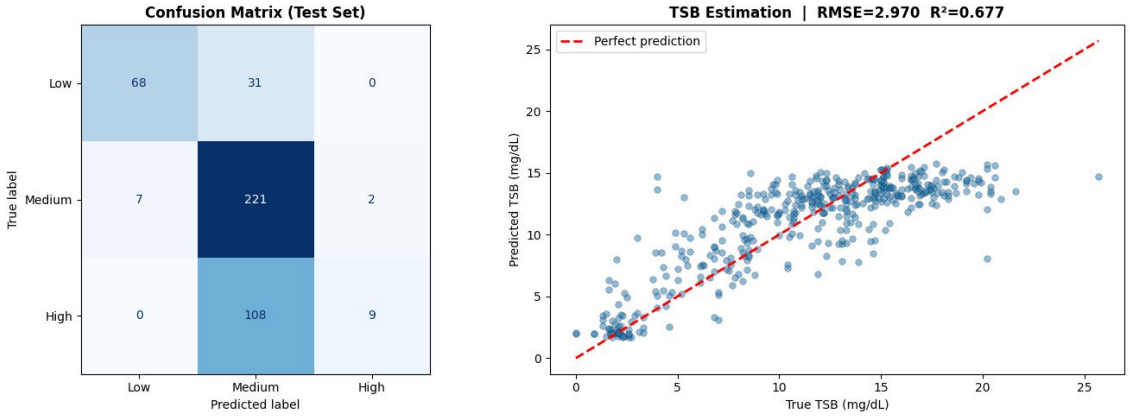


Figure 6.6: Performance evaluation of the 2D-CNN model: (Left) confusion matrix across Low, Medium, and High risk categories; (Right) TSB regression plot ($R^2 = 0.677$, RMSE = 2.970 mg/dL).

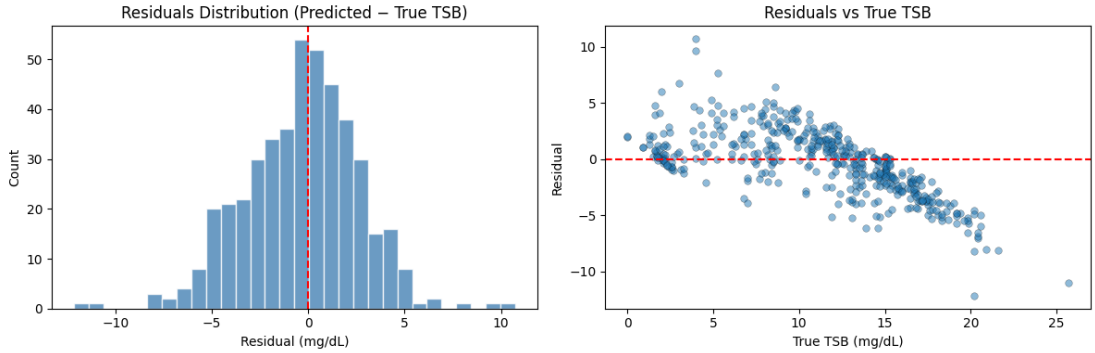


Figure 6.7: 2D-CNN residual analysis: (left) residuals distribution histogram showing near-zero mean bias, (right) residuals vs. true TSB scatter showing heteroscedastic spread above 15 mg/dL.

Result: 72% accuracy. The lower result, despite a far larger parameter budget, is explained by the domain gap between ImageNet's natural image features and the close-up, colour-calibrated, skin-only ROI patches in our dataset. ImageNet pretraining optimises for distinguishing hundreds of diverse object categories; those low-level colour representations are poorly aligned with the subtle bilirubin-induced colour shifts in neonatal skin.

6.7 Model 4: ResNet-50 with Full Preprocessing Pipeline

ResNet-50 was trained on images produced by the fully corrected preprocessing pipeline, including HSV colour card detection, LAB white balancing, yellow-region detection, and morphological ROI extraction.

Result: 78% accuracy. Unlike the 1D-CNN result of 76%, per-class precision and recall confirmed this figure reflects genuine multi-class discrimination rather than class-prior exploitation. The improvement over EfficientNet-B0 is attributable primarily to the improved



Figure 6.8: 2D-CNN + MLP fusion model: training vs. validation loss across 5 cross-validation folds. All five folds show rapid initial descent followed by stable convergence, with validation loss closely tracking training loss.

preprocessing.

6.8 Comparative Summary

Table 6.1: Comparative Performance of All Model Configurations

Model	**Accuracy**	**Input Type**	**Key Finding**
1D-CNN (baseline)	76%	Image $\rightarrow$ 1D	Variable sequence length; class-imbalance artefact
1D-CNN + MLP (fused)	75% / R^2=0.687	Image + GA, PNA, BW	Clinical fusion improves regression; best MAE 2.42 mg/dL
2D-CNN (optimised)	81.33%	567×567 ROI	Mode collapse resolved; cross-dataset transfer observed
2D-CNN + MLP (5-fold CV)	Stable across folds	567×567 ROI + clinical	Consistent generalisation; low variance across folds
EfficientNet-B0 (TL)	72%	567×567 ROI	Domain gap with ImageNet features
ResNet-50 + Full Pipeline	78%	567×567 ROI	Genuine multi-class learning confirmed

6.9 Key Observations and Lessons

- **Preprocessing dominates architecture.** The performance gap between the 1D-CNN (76%) and ResNet-50 with full preprocessing (78%) owes more to the improved input pipeline than to the architecture change.

- **Clinical fusion consistently improves regression.** Across all 1D-CNN variants, adding gestational age, postnatal age, and birth weight lifted R^2 from 0.546 to 0.687 — an improvement that no architectural change alone produced.

- **Class imbalance is a first-class problem.** At 81% majority-class prevalence, raw accuracy

is not a valid primary metric. F1-score and ROC-AUC are the appropriate indicators.

- **Domain gaps are real.** Transfer learning from ImageNet underperformed a smaller custom CNN on this domain-specific task.
- **Colour calibration enables cross-dataset transfer.** The 2D-CNN model generalised meaningfully to the private corpus despite differences in imaging hardware and lighting.

CHAPTER 7

LIMITATIONS AND FUTURE ENHANCEMENTS

7.1 Current Limitations

7.1.1 Dataset Size and Diversity

The NeoJaundice dataset contains 745 unique neonates, which is modest relative to the diversity of skin tones, lighting conditions, camera hardware, and clinical contexts a deployed system would encounter. Models trained on this dataset may not generalise well to populations with very different skin tone distributions.

7.1.2 Regression vs. Classification

The current framework operates in classification mode. Continuous bilirubin regression against TSB values in mg/dL provides richer clinical information but requires more careful evaluation methodology and was not the primary output in this iteration.

7.1.3 No Real-Time Deployment

The current implementation runs in a research notebook environment. A deployable mobile application has not been built, and the computational requirements of the full preprocessing pipeline have not yet been profiled on mobile hardware.

7.1.4 Interpretability

The model produces a risk probability but provides no spatial explanation of which skin regions drove the prediction. Clinicians and regulators will require this level of interpretability before any clinical deployment.

7.2 Future Enhancements

7.2.1 Ablation Studies

A formal ablation study quantifying the independent contribution of each component — colour calibration, yellow-region detection, clinical parameters — is the first planned follow-up.

7.2.2 Grad-CAM Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) visualisations will produce spatial attention maps showing which skin regions the model attended to [23], directly addressing the interpretability gap.

7.2.3 Mobile Deployment

TensorFlow Lite model quantisation and pruning will be explored to reduce the model footprint for deployment on Android or iOS devices.

7.2.4 Larger and More Diverse Datasets

Collaboration with clinical partners to collect a larger and more demographically diverse dataset — particularly covering darker skin tones and lower-resource clinical environments — is the most impactful single enhancement available.

7.2.5 Continuous Bilirubin Regression

Extending the framework to output a continuous TSB estimate rather than a binary risk label would enable bilirubin trajectory monitoring over time, rather than point-in-time screening only.

CHAPTER 8

CONCLUSION

The basis of our project consisted of a quite obvious statement: children are born with jaundice, the detection of which requires a painful needle stick procedure that has to be done since the blood is the only option to diagnose this disease. However, the current alternatives to the blood test are way too expensive to be used by those who need them the most. Therefore, the main issue raised as the motivation for our project concerned whether it was possible to create something combining a camera and three pieces of information regarding the newborn child, which would replace blood sampling.

There are three major successes of our project. First, a set of preprocessing algorithms pipeline that turned out to be quite useful for increasing the effectiveness of neonates' skin images representation; color normalization was revealed to be the most effective preprocessing step. Secondly, our experiments proved that a correct choice of preprocessing steps is more important than using some kind of particular models when predicting bilirubin concentration. Lastly, the experiment proved that the fusion of gestational age, postnatal age, and weight increases the regression R^2 – from 0.546 to 0.687

Whereas the issue of class imbalance when the majority class accounts for 81% was found to be the greatest difficulty in comparison to any other design choice. Although mode collapse was a real problem that was addressed by the appropriate means, it must be stated that, considering the medical field in question, it is more dangerous to have a false negative result rather than false positive. Thus, all necessary measures should be taken.

The main point to learn from this entire project experience is the necessity to integrate clinical reasoning into the machine learning architecture from scratch when working on a machine learning solution for a particular medicine-based application. It goes without saying that a neonatologist looks beyond the color of the skin of the child in his care.

This seems to be a realistic attempt and should be attempted where a midwife takes the picture of the two-day-old infant, enters her/gestational age, post-natal age, birth weight, and gets the bilirubin risk assessment result.

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